Supplementary Material for "From Embodied Experience to Reusable Relational Knowledge: An Auditable Unified-Graph Architecture and a Preregistered Boundary Analysis of Activation-Based Access"

Kailin Yan

S1 Terminology, Evidence Levels, and Protocol Layers

Terminology. *Experience*: a temporally ordered action–observation stream with timestamps and settlement outcomes. *Relational knowledge*: typed assertions encoded as graph nodes and typed, weighted directed edges—distinct from raw log entries and vector embeddings. *Write-back*: the pipeline step that converts experience fragments into candidate relations and promotes them into persistent structure; it is the *only* pathway from experience to knowledge in FAS. *Persistent*: structure survives session/episode boundaries, with runtime load (activation values) inherited along with it. *Reuse*: existing structure relevant to a new situation is re-lit and enters the cognitive working set. *Functional consumption*: stored knowledge enters action selection through an explicit consumer mechanism—not "knowledge exists" but "knowledge changes behavior."

Production and experimental paths. To avoid confusion, all evidence is attributed to four layers, labeled at each occurrence: **P1 production default path**—zero-LLM subject behavior (graph, activation, ledger, modulation, intention loop, replay gate) runs with production configuration in the sandbox. **P2 production mechanism, assembled validation**—ledger (Exp. C) and event framework (`enabled=False`): the mechanism lives in production code; differences are exposed by experimental assembly. **P3 mechanism-isolation layer**—the core incremental-value campaign and the relation-direction diagnosis: controlled mechanism experiments on reference models distilled from production semantics; no behavioral benefit in the production loop is claimed or required. **P4 experimental-layer harness**—the LLM decision head (not attached to production).

Two assembly audits are reported as such. (i) An early ablation matrix ran on an assembly missing the diffusion executor (the continuous-cognition loop was not wired), so its zero differences only support "completion-level insensitivity to activation"; after repair, access-layer experiments carry the mechanism evidence. (ii) The first routing harness never built edges, swallowed 521/521 exceptions, and always produced an empty first-step focus—it was declared invalid and rebuilt and re-run (Section S11); a later audit additionally found that the production answer pipeline originally did not render attention/demand annotations, and it has since been wired to the same cognitive structure, aligning production and harness paths on annotation rendering—this alignment fixes the reporting definition but changes no experimental number. Combined constraint: mechanism- and assembly-level evidence fully supports this paper's claims, but no "production superiority" statement may be extrapolated from it.

S2 Internal Code-Name Mapping

Historical run-family codenames appear in archived filenames and statistics files. Table S2 maps them to the neutral names used in the paper; the main text avoids them.

Table S1: Evidence levels × protocol layers (full merged scale; Table 1 of the main text is the abbreviated version).

Level	Meaning of a claim at this level	Layer the evidence comes from
L1 representational	learned structure exists and re-activates	P1 (production default path, zero-LLM)
L2 mechanism	measurable activation/ledger dynamics	P2 (assembled validation) and P3 (mechanism isolation on reference models)
L3 behavioral	stored knowledge changes action selection	P2 consumer assemblies; P4 decision-head harness (exploratory)
L4 general capability	—	*explicitly not claimed anywhere*

S3 Experiment Families and Archival Census

Every run is archived by the run recorder (metadata, event stream, start/end graph snapshots, metrics) into content-addressed directories. The archival census of behavioral families totals 494 runs (ablation 77, eventframe 53, sandbox 43, ci 34, transfer 32, reuse_a2 31, prior_boundary 31, longchain 29, reuse 28, emotion 25, hebbian 24, event_transfer 22, diffusion_b2 22, envchange 14, diffusion 12, baselines 8, prior_conditions 7, minecraft 2); further, the mechanism campaign comprises 34,016 runs (line-countable in its raw results file), the beacon campaign 80, the routing campaign 520, the relation-direction diagnosis 60, and the falsification campaign 663; the shared-hub parameter sweep (Section S8) adds 2,550 runs. Failures and pilots are all retained.

Provenance notes. (i) The mechanism campaign's primary numbers use the independent 663-run falsification campaign's statistics as the reference-model source; a directory named `experiments/mechanism_campaign` is a *different*, later activation-landscape audit whose probes failed and whose formal runs never executed—it is not a number source. (ii) The routing campaign's first statistics file was renamed with an INVALID suffix and documented; the current statistics file is the official v2 campaign.

S4 Component Status

S5 Write-Back: Worked Example, Taxonomy, and Claim Audit

The full promotion list of the write-back-on lineage (15 relations) and the corresponding graph snapshots ship with the artifacts. Example promotion (seed 1): `gather_resource(oak_log)` leads-to `change:inventory:oak_log:count_increased`, cumulative support 8 / confidence 0.889 after 8 rounds, both endpoints mark-activated on write.

S6 Beacon Task: Case Study and Embedding-Baseline Check

What kind of retriever is D-flat? D-flat is a genuine sentence-embedding retriever: it encodes all graph node identifiers with a public, locally runnable sentence-transformer model (BAAI/bge-small-zh-v1.5) and ranks them by cosine similarity to the task text—no activation, no propagation, no path structure. We verified this at import level (the retrieval path instantiates the embedding provider and computes normalized cosine similarities; no diffusion call is reachable from it).

Zero-LLM accessibility check. Design (frozen before execution): rebuild the beacon step-0 assembly for each of the 20 paired seeds (production graph-building path with the recipe closure; no observations ingested), compute the full-graph embedding similarity ranking for the exact goal text used in the campaign, and record the ranks of the recipe and resource nodes. Result: the graph is identical across all 20 seed assemblies (44 nodes; deterministic construction); `item:oak_planks` ranks **2nd** and `item:oak_log` ranks **4th** in every assembly—both inside the Top-8 context window that D-flat presents to the decision head. This is the enabling condition of D-flat's 20/20 result, verified without any LLM call. (Purely descriptive; no statistical test applies to a single deterministic assembly state.)

Table S2: Internal code-name mapping (for navigating the archives; not used in the main text).

Internal code	Neutral name	Note
A10	beacon task	information-asymmetry closed-book task (main text Section 5.1)
B0/B1/B2 (writeback)	write-back isolation lineages	B2 here = write-back-on lineage; *do not conflate* with B2 below
B2 (core incremental value)	flat path-product baseline	closed-form flat baseline of the mechanism campaign; distinct meaning from the writeback lineage
B2 (diffusion_b2)	four-condition access family	a third use of the label, historical
B1 (core incremental value)	direct injection baseline	graph-free control
B3 / B4	random activation / oracle	equal-energy control; generator-correctness precheck
C-1..C-4	preregistered conditions 1–4	the four confirmatory topologies of the mechanism campaign
O1–O3	topology detectors O1–O3	O3 = shared-hub (combinational) topology
D-flat / D-noact	embedding-retrieval / activation-off controls	beacon and routing conditions
D (event framework)	event-framework experiment	assembled validation of the production-default-off channel
C20e / C20g	event-framework follow-ups	real-failure-receipt variants
C-emo	affective dose experiment	scalar modulation dose response
D-CI	intention lifecycle experiment	continuous-intention formation through decay
F (hebbian)	Hebbian isolation	weight-level isolation, behavior sub-threshold
G0/G1/G2	re-activation conditions	full graph / episodic-stripped / clean
H/K/L (longchain)	long-chain variants	L0 full, L1 −diffusion, L2 −write-back, L3 −priors, L3b prior restored
I/U (envchange)	environment-change family	scale insufficient; re-validation probes
J/P (prior_boundary)	prior-level family	P0 zero priors, P1 mechanics tables only, P2 full
M1–M3 (minecraft)	live sessions	observational only
T1–T4 (campaign)	mechanism topologies	3in/100n/25% to 5in/1000n/75%
T1–T4 (routing)	decision-head task classes	knowledge reuse / interference / re-routing / competing targets
I1–I3 (relation dir.)	injection conditions	single, single, dual input
F / B (relation dir.)	forward / bidirectional projection	production graph vs. +reverse-edge projection
A/B closure	closure-masking checks	completion-layer insensitivity instances

S7 Activation Mechanism: Access-Layer Separation

Four-condition access task (n=5, deterministic). On an access task—diffusion on; diffusion off (β=0, injection only); random-activation control (same energy injected into random semantic nodes); retrieval-oracle control (target-chain nodes raised every tick)—full diffusion produces activation quality 61.5 over a field of 39 lit nodes; diffusion off 16.6 over 17; the random control is comparable in quality (45.8) but with a larger, structureless field whose out-of-sight targets light at drift moments (ticks 1–9 across seeds) rather than in a fixed pattern; the oracle 17.2 over 18 (Fig. S2). Structural diffusion is therefore distinguishable from both noise and lookup at the knowledge-access layer.

Companion causal experiment (re-run after the assembly repair). Gap "pointed-at" entity activation 1.5→5.0 capped (control wanders 1.1–1.7); *out-of-sight* pure-diffusion signal `oak_log` 0.43 vs 0.0; gap-source energy drains under emit-as-transfer (1.43→0.16) while the control sticks; 2-hop lit field 5 vs 4. Completion layer: zero difference in both experiments (closure masking; main text Section 5.4).

Mechanism-campaign ancillary results. B1 direct injection and B3 random activation confirm the contrast setup works independently of the graph; the B4 oracle precheck (MRR= 1.0 on all four topologies) excludes generator-side error. Hidden oracle (30 seeds, label/ID permutation): target rank 22.5 unchanged. Insertion order: 10 permutations, CV= 0.0. Counterfactual C2 (no joint target): all four methods 0.0. Negative-edge test: joint top-1 0.0333=0.0333, total score 17.41 unchanged (no collapse).

Per-condition confirmatory numbers (preregistered, n=50 paired seeds each). Median ΔJCG: −0.402 (C-1), −0.402 (C-2), −0.784 (C-3), −0.782 (C-4); reverse-direction one-sided Wilcoxon with Holm correc-

Table S3: Family → paper-section provenance.

Family (internal code)	Main-text section	Purpose / headline number
Beacon (A10)	§5.1	information asymmetry: 20/20 vs 0/20 vs 0/20 vs 20/20
A2 / reuse	§5.2	8-round promotion of 4 relations per seed
B0/B1/B2 (writeback)	§5.2	write-back isolation 0 vs 13 nodes
G0/G1/G2	§5.2	cross-context re-activation 4/12–0/8–0/0
Exp. C (transfer)	§5.2	ledger channel 13 vs 16 ticks
Exp. D / C20e/C20g	§5.2	event-framework channel @6 vs @9, @15 vs @17
Exp. B (diffusion)	S7	access-layer separation (post-assembly-fix)
B2 (diffusion_b2)	S7	four-condition separation 61.5/16.6/45.8/17.2
falsification	§5.3	663 runs: JCG> 0 rate 0.970, hypothesis-group MRR@1= 1.0
core incremental value	§5.3, S7	34,016 runs: C-1..C-4, combo, sweeps, counterfactuals, oracle
shared-hub sweep (new)	§5.3, S8	2,550 runs: 84-cell grid + O3 replication
C-emo (emotion)	§5.4	dose response 49.8/52.5/57.4
D-CI (ci)	§5.4	intention lifecycle 3–4 per run; expression unattained
F (hebbian)	§5.4	Hebbian weight isolation +0.074; behavior sub-threshold
H/K/L (longchain)	§5.4	315 / 259 / 0-of-5 / 315 ticks
J/P (prior_boundary)	§5.4	P0 0/25; P1=P2 25/25
I/U (envchange)	S12	environment-change scale insufficient
M1–M3 (live)	§5.4	observational forms; live causal ledger
A/B matrix (ablation)	S1	completion-layer masking
routing v1 (invalid)	S11	declared invalid; not a basis for claims
routing v2	§5.4, S11	520 runs, exploratory boundary
relation direction	§5.3, S9	direction-semantics mechanism diagnosis
determinism re-run (new)	§5.5, S10	20 re-runs: outcomes 20/20, trajectories 7/20

tion p=1.5 × 10^{-9} (the frozen implementation's one-sided p is half the two-sided p regardless of direction; the preregistered-direction tail is its complement, ≈1; see main §4.2); effect r=0.615; seeds where FAS beats flat: 0/200. MRR (FAS vs flat path-product): 0.0435/0.0426 vs 0.5/0.5 on the two smaller topologies; tie at 1.0 on the two larger. Robustness sweep: slope 0.0076 (flat 0.0), AUC 0.0427 vs 0.04, critical noise 0.0=0.0. Weight sweep: both methods reach MRR= 1.0 together at w=0.7 (FAS 0.083@0.5 → 1.0@0.7; flat 0.077 → 1.0).

S8 Shared-Hub Parameter Sweep: Design and Full Data

Design (frozen before execution). Question: where in (edge weight w, converging-input count k, private-path depth d) does activation-based access beat the closed-form flat baseline on the shared-hub structure? Topology per cell and seed: k inputs, each connected by d private hops (edge weight w) to a shared hub, one hop (w) to the joint target T; each input also has a 2-hop private path (weights 0.8/0.8, fixed, mirroring the original shared-hub design's branch strength) to a single target S_i; each input carries 4 two-hop distractors (weight 0.3), 2 of them in the candidate ranking domain; candidates = $\{T\} \cup \{S_i\} \cup$ distractors. Grid: $w \in \{0.3, \ldots, 0.9\}$ (7) × $k \in \{2, 3, 4, 6\}$ (4) × $d \in \{0, 1, 2\}$ (3) = 84 cells, 30 paired seeds per cell; plus an exact replication cell of the original shared-hub configuration (k=2, d=0, w=0.5, no distractors) as a hard preflight. Diffusion parameters identical to the preregistered campaign (steps = joint hops +2; seed strength 5.0; ratio 0.5; emit-as-transfer 1.0; decay 0.05; cap 5.0). Metrics: success@1 on T (primary), MRR, JCG. Statistics: per-cell paired exact McNemar (success) and two-sided Wilcoxon (MRR); the sweep is *descriptive*—no universal multiple-comparison correction is applied, and no single-cell p is used confirmatorily; the paper cites only sign-consistent cell counts and the surface shape. Failing-test criterion (declared in advance): if no cell shows FAS strictly better, the paper's wording reverts to "the single-cell positive difference found no parameter-region support." A design correction (v1.1, before any formal run): d was initially mis-specified, which made the original configuration unreachable in the grid; the preflight failed as designed, d was redefined as private-hop count with range $\{0, 1, 2\}$, and the corrected

Table S4: Component status (implementation / experimental validation / evidence level / default switch).

Component	Impl.	Validated	Level	Notes / default
Unified graph (single graph, 7 label classes, 8 relation types, negative weights)	production	all experiments	L1	auxiliary layers never build structure
Spreading activation (decay/gain/suppression/Top-k/emit-as-transfer)	production	access tasks, four conditions	L2	runs every tick
Evidence-weighted write-back (window/support/confidence/promotion)	production	isolation, reuse, event exps	L1/L2	sole structure-writing channel
Persistence and inheritance (structure + activation)	production	re-activation, ledger exps	L1/L2	cross-episode; snapshot rotation + shrink guard
Ledger consumers (reward ledger + trial ledger)	production	transfer exp	L3	prior bonus ± 0.05
Event framework (polarity-slot edges)	implemented	event exps (assembly)	L2/L3	**production default OFF**
Affect/neuromodulation (graph citizens: tonic truth, derived level, graph as effect source)	production	dose exp	L2	every tick
Hebbian co-activation	production	isolation exp	L2	default on; behavior sub-threshold
Intention lifecycle	production	lifecycle exp	L2	expression level unvalidated
Offline experience replay gating	production	not touched in this paper	—	idle gating
Episodic buffer (capacity/forgetting/retrieval/promotion)	production	not touched in this paper	—	short-horizon integration layer; promotes tags only
Web-search action node	production	not touched in this paper	—	external knowledge acquisition
Hormone graph citizenship (tonic real value, derived level, graph as effect source)	production	not touched in this paper	—	hormone *direct behavioral* control not implemented
LLM decision head	experimental	routing campaign, 520 runs	L3 (exploratory)	not attached to production
Prospective memory, full perception pipeline, symbolic reasoning, RL policy nodes	not implemented	—	—	future work

design was frozen before any data was produced.

Results (2,550 runs). Preflight replication: FAS 30/30 vs flat 0/30 (McNemar $p=1.9 \times 10^{-9}$)—the original single-design finding reproduces exactly. Grid: FAS strictly better in **11/84** cells, flat strictly better in **9/84**, tie in 64.

On the original-configuration axis (k=2, d=0): FAS/flat success 0/0 at w=0.3, 8/0 at 0.4, 30/0 at 0.5, then 30/30 at $w \geq 0.6$—flat catches up exactly when the hub-path weight reaches 0.6 (median MRR: FAS 0.33/0.50/1.00 vs flat 0.33/0.33/0.33/1.00 across w=0.3–0.6). All 84 cell-level rows, raw per-seed metrics, and the extraction script ship with the artifacts; the numbers in this section and the main text are read programmatically from `sweep_summary.csv` / `paper_numbers.json`.

S9 Relation-Direction Diagnosis: Full Data

Static audit (production recipe graph, 44 nodes/53 edges): the five relation classes (produces 8 / belongs 18 / drops 13 / needs 11 / needs-tool 3) have all-sink targets (sink fraction = 1.0); `item:*` 13/13 with zero

Table S5: Write-back-lineage (seed 1) 15 promoted-relation taxonomy. Classification by edge content; the two window-polluted pairs are a subset of the substantive class.

Promoted relation	Class	Reading
`gather_resource(oak_log)` → `change:inventory:oak_log:count_increased`	substantive	mining yields items
`gather_resource(oak_log)` → `change:block:oak_log:disappeared`	substantive	mining removes blocks
`craft_item(oak_planks)` → `change:inventory:oak_log:count_decreased`	substantive	crafting consumes logs
`craft_item(oak_planks)` → `change:inventory:stick:count_decreased`	substantive	crafting consumes sticks
... (7 further inventory/block pairs)	substantive	
`gather_resource(oak_log)` → `change:inventory:oak_planks:count_increased`	substantive, window-polluted	downstream craft attributed to upstream gather
`gather_resource(oak_log)` → `change:inventory:stick:count_increased`	substantive, window-polluted	same
$\{A\}$ → `change:self:{A}:succeeded` (×4)	degenerate	action → own success; tautology

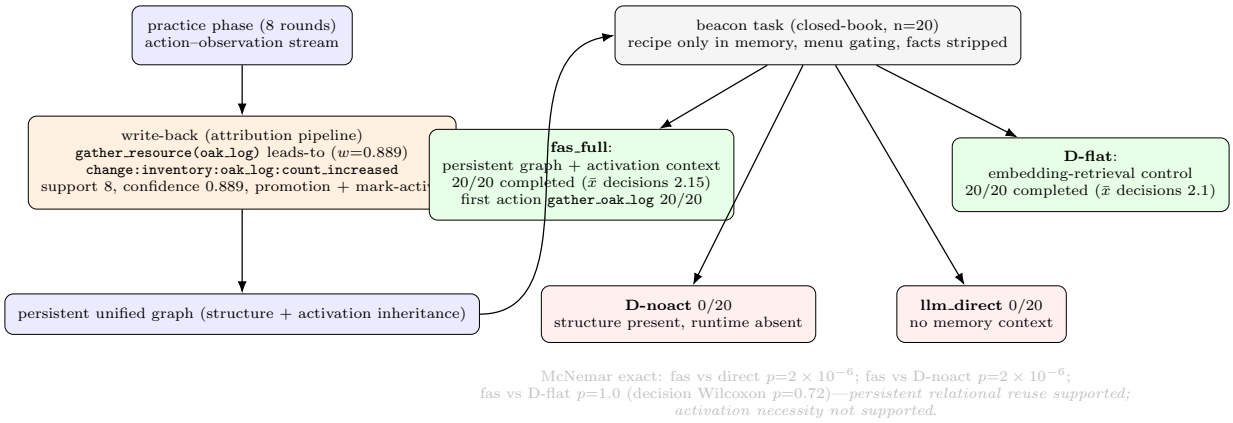


Figure S1: A complete real case study: the wood-to-plank chain from practice experience, through evidence-weighted write-back into persistent graph structure, to the behavioral benefit in the information-asymmetric beacon task. All numbers from the archived beacon analysis (80 runs, four conditions × 20 paired seeds).

out-edges; a depth-1 star forest; forward 2-hop reachability 0. Construct graph B = production graph F + 41 generic reverse projections (53→94 edges, no node special-casing; sandbox-memory only, production graph zero modification). 60 runs = 2 graphs × 3 injection conditions (I1/I2 single input, I3 dual input) × 10 paired seeds. All 21 paired tests give p=0.001953—the minimum attainable two-sided p at n=10 (exact Wilcoxon, all seeds concordant), r=0.886, 10/10 seeds in the same direction; this certifies uniform direction, not effect size.

Main-axis numbers: forward 2-hop reachability F always 0, B 10/9/12; 3-hop B 6; lit nodes F 6/6/9 vs B 22/21/27; rank correlation between topological score and diffusion activation ρ F 0.265/0.275/0.325 vs B 0.542/0.513/0.645 (dual input: p=0.002); shared-node (`item:stick`) absolute activation (dual input) F 0.833 vs B 0.475 (single-input 0.417 vs 0.253/0.239); additive prediction 0.491 ≈ measured 0.475; Gini 0.417 → 0.792–0.824.

Not run, recorded as such: the suppression-off control is unavailable because the architecture has no global negative-weight switch; no behavioral experiment this round; the production graph was not modified.

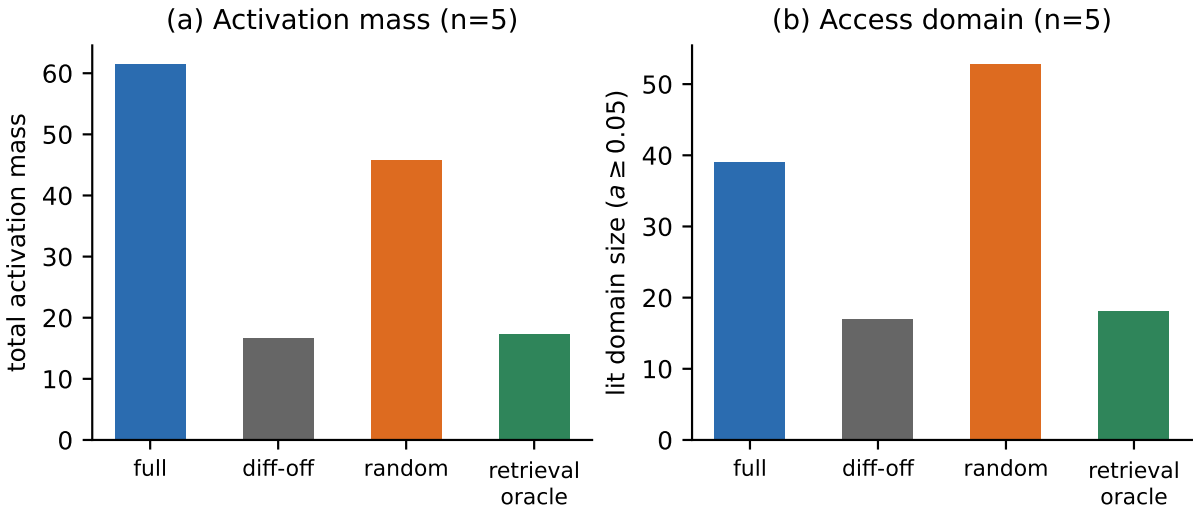


Figure S2: Four-condition activation task (n=5, deterministic). Structural diffusion produces a distinct access field; same-energy random activation cannot reproduce it; the target-chain oracle is narrower.

S10 Determinism Re-Run of the Decision-Head Family

Design (frozen before execution): re-run a 5-seed × 4-condition subset of the beacon campaign (seeds 0–4; conditions fas_full / D-noact / D-flat / llm-direct; identical code path, fresh output directory) and compare against the archived campaign runs on success, decision counts, steps-to-craft, first action, and the full action sequence.

Reading: task outcomes reproduce exactly, but trajectories do not—the remote LLM endpoint is not bit-deterministic even at temperature 0 (provider-side state is outside our control). The paper therefore claims trajectory-level determinism only for the zero-LLM families; the decision-head families are outcome-robust but not trajectory-reproducible, and no claim rests on their exact trajectories.

S11 Routing Campaign: Full Results

v1 invalidation record. The first campaign was rejected by assembly audit before formal writing: (R1) the experimental graph had zero edges (the runner never built edges; activation only traveled existing edges); (R2) 521/521 demand/gap/routing calls threw and were swallowed into placeholder text (the "−demand" ablation was an identity); (R3) 80/80 first-step focus empty. No v1 number (including anything in earlier paper drafts) is a basis for any claim in this paper; the stale statistics file was renamed with an INVALID suffix and documented, and the current statistics file is the official v2 campaign.

v2 results. The rebuilt official campaign runs on the production edge-building path: seeding a real graph (37 nodes), real observation-driven edge construction, core exceptions propagate, and 9 harness invariants are fail-fast; $5 \times 4 \times 20 + 3 \times 2 \times 20 = 520$ runs in total, all valid. Token statistics (per-decision prompt means): FAS context 1456/1084/2070/2662 (T1–T4) vs no-context 661/528/554/1267, i.e., 2.1–3.7×; all 20 principal token comparisons $p \leq 0.003$. Mechanism telemetry (ablation family n=40): diffusion ON raises activation divergence 2.4→9.9, pick precision 0.717 vs 0.734, demand annotations present 100% vs 0%.

Reading. After Bonferroni correction over the campaign's 280 tests, all three competing-targets metrics give $p \approx 1$; that task is directional only. The positive reading is "FAS context ≠ no context" in an information-competition scenario; mechanism telemetry joined with the shared-hub analysis implicates associative recall on observation-co-occurrence subgraphs, not propagation of recipe knowledge, as the carrier. The re-routing efficiency deficit and token costs are the robust cost side. The campaign's boundary assumption (fully observable tasks ⇒ limited marginal value of memory context) is stated in the main text, together with its untested complement (information asymmetry)—the beacon task provides the first closing evidence for exactly that complement.

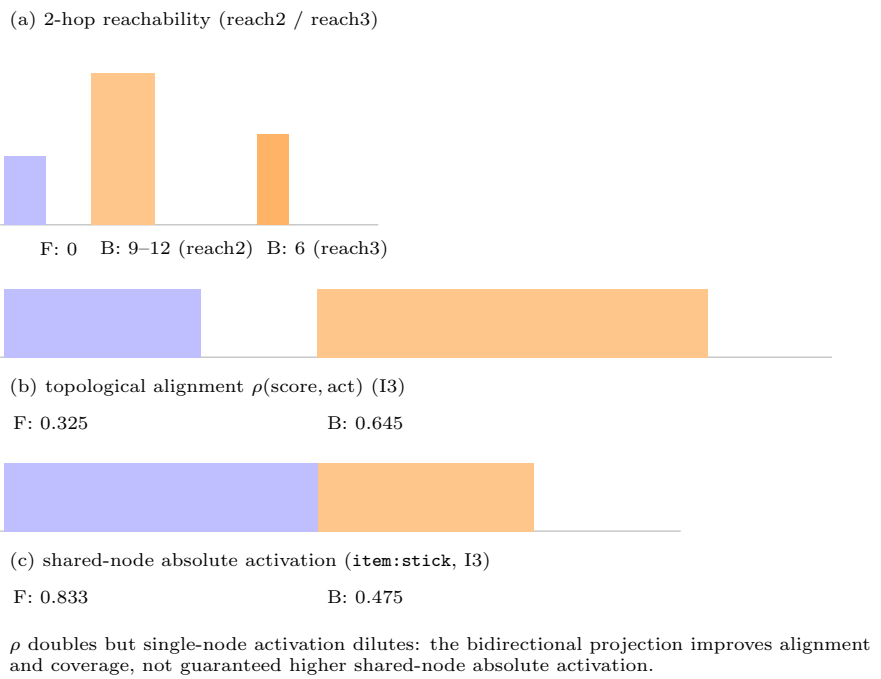


Figure S3: Relation-direction experiment summary (n=10 paired seeds; all 21 tests p=0.001953, r=0.886, 10/10 seeds same direction). (a) bidirectional projection restores multi-hop reachability; (b) topological alignment doubles under dual input; (c) shared-node absolute activation is instead lower—the benefit comes with a dilution cost.

S12 System-Level Details

Long chain. A 14-step chain (13 in-recipe steps plus the documented smelting-instant-settlement simplification, a two-tool intermediary branch, decoy resources) completes 5/5 seeds at 315 ticks under full configuration (L0). Removing graph prior edges *and* the smelting task prior (L3): 0/5 (65 actions / 58 redundant / 33 failed)—the agent cannot plan out-of-recipe transitions. Re-adding only the smelting prior (L3b) restores the bit-identical 315 ticks. −diffusion (L1) is faster (259 ticks); −write-back promotion (L2, 0 promotions) equals L0 (315 ticks)—completion is insensitive to knowledge-organization differences.

Zero-prior starvation. Three prior levels: P2 (full priors), P1 (mechanics tables only), P0 (zero priors); two-step craft task, five episodes per seed. P1 and P2: all 25 episodes complete (median 4 ticks, indistinguishable). P0: **0/25**. Direct replay diagnosis: under P0 the closure planner finds no path to the target's product, the goal system degrades into perpetual `explore_area(target)`; after 60 ticks the inventory is empty and the attribution ledger is completely untouched (0 aggregations / 0 hypotheses / 0 promotions)—though the graph still grows from perception (+29 nodes over 5 episodes). Zero priors do *not* constitute a small-sample test of learning ability: the agent already failed at upstream experience generation.

Environment change. The environment-change family's scale proved insufficient for the intended re-validation questions (recorded as such); its re-validation probes (Q1–Q3) confirmed assembly integrity.

Live observations. Real Minecraft bridge (zero LLM in the action path): M1 (25 min, discovery form) net graph +100 nodes, +365 edges, 400 events (a recorder defect lost 86, patched and labeled as such); M2 (25 min, convergence form) only +35 nodes/+95 edges with dominant behavior shifted to learned goals (investigate 89); the live causal ledger (89 aggregations / 78 hypotheses / 29 promotions) comes entirely from real behavior. The causal bridge "write-back → behavioral improvement" is not established in the live window (M2 had no obtain closed loop—an honest negative observation).

Baselines family. Reactive, memory-only, graph-retrieval, and LLM-heavy baseline assemblies exist in the archives as no-cognition controls for the subject experiments; they appear in the closure-masking checks (all complete identically on closure-guaranteed tasks).

S13 Configuration and Reproducibility

Sandbox clock 16 s/tick; attribution window 8–120 s; promotion thresholds $s \geq 5$, conf ≥ 0.80 (Laplace $s/(s+c+2)$); diffusion clamped by per-space caps (episodic 3.0 / global 5.0), category-dependent λ, emission normalized by positive weight, emit-as-transfer coefficient 1.0 (sources exempt); modulation clip $[0.7, 1.6]$; Hebbian ϵ=0.0015, $\gamma_{\max}$=0.5, whitelisted classes {`semantic-relation`, `emotional-relation`}; intention cap 3, formation 0.55, expression 0.78, discard 0.10, cold-field threshold 0.35; event-framework production switch `experience_eventframe.enabled=False`; beacon hooks: closed-book recipe, craft-menu gating, facts stripped, goal seeding off; mechanism campaign: T1–T4 topologies (3in/100n/25%, 3in/300n/50%, 5in/300n/50%, 5in/1000n/75%), B1–B4 generators, JCG definition, Holm correction, committed seed reservoir (full machine-readable configuration ships with the artifacts). All regression checks pass at the time of writing: 82 unit tests in the pytest suite and 108/108 checks in the regression-script runner.

Artifacts. The repository ships: (i) the deterministic sandbox and production stack; (ii) all archived campaign outputs (raw run records, graph snapshots, statistics files); (iii) one-click scripts for the zero-LLM replications (mechanism campaign, shared-hub sweep, embedding accessibility check); (iv) the preregistration records; (v) a claim-to-evidence mapping (Section S14). Decision-head replications require the (anonymized) LLM endpoint configuration and reproduce outcomes, not bit-identical trajectories (Section S10).

S14 Claim → Evidence → Source Mapping

S15 Capability-Gap Campaign v2: Protocol and Tables

A second, baseline-controlled campaign (508 archived runs; design frozen in `experiments/capability_gap_v2/`) tested the persistent state against LLM baselines at the behavioral layer. Two mid-course confounds were caught and repaired with full archival separation (gate rule v1 → v2; P3 inventory carry-over), and a self-model ablation turned out to hook a nonexistent method and was rebuilt with telemetry verification (Table S12).

The resumption advantage is regime-dependent (short-history streams only). Full protocols, confound repair logs, and the fault classification table ship with the campaign archive.

S16 Persistent-State Advantage and Controls: Tables

Interpretation: on T2, no difference is detected between FAS-full and the structure-preserving controls (10/10 each, n=10, limited power)—persistent goal context is what matters, whatever its carrier. On T3, C1 repeats less than C3/C4 (Bonferroni-significant), but the post-hoc audit traced this difference to failure-evidence visibility: the control implementations structurally excluded failure information from the decision context (and the naive ranking rule placed failure facts last), so the independent contribution of graph structure or spreading activation is not established here. A frozen-preregistration revision (flat failure-first; 2-hop adjacency; 2-hop without polarity edges; `persistent_state_controls_revision`) was gate-stopped at its presentation-equivalence precheck, before any LLM run: the flat failure-first control passed the gate (failure fact at slot 6, char ratio 0.753 vs C1), but under the naive 2-hop rule the failure-evidence node is structurally reachable (51-node 2-hop set) yet ranked 43rd by discovery-edge weight (0.5 vs a slot-8 weight of 0.8) and truncated out of the 8-slot budget (char ratio 0.349). Failure-evidence absence in graph controls is jointly determined by distance, ranking weight, and budget; information equivalence does not imply presentation equivalence; the preregistered stop rule took effect with zero runs, and the revision's hypotheses (H_R1–H_R4) remain unanswered. See `PREREGISTRATION.md`, `ADDENDUM_T3_AUDIT.md`, `AUDIT_REPORT.md`, and `persistent_state_controls_revision/` in the archives.

Run accounting. The campaign-v2 archive separates official batches (508 runs: A 55, B 40, C 30, E 70, F 30, I 102, J 10, K 83, T 54, T2 30, T-mech 4) from repair/confounded batches (390 runs: C gate artifact 40 + C open-facts 30, E pre-repair 30, F repair variants 210, I gate artifact 39, K design-inert 36, T mock 5). Headline campaign numbers cite only official batches; repair batches appear only as audit history (the two mid-course confound repairs) and are never pooled with official data. The persistent-state archives contain no excluded runs: FAS-sham reaches 10/10 on T2 with all ten seeds valid (early internal notes citing 9/9 were a mid-run progress snapshot, not a validity exclusion).

S17 Statistical conventions: full table

Table S15 is the full version of the main paper's compact Table 8: every comparison family with its test, sidedness, pairing, multiplicity handling, status, and the archived implementation that produced it. All values are reproducible from the package (`scripts/reproduce_key_numbers.py`, steps 1–5); the ΔJCG family is additionally reproduced bit-exactly by the frozen implementation (step 3b).

Table S6: Implementation → claim audit (excerpt; the full 17-claim audit ships with the artifacts). Verdicts: S = mechanism exists and is experimentally verified; Q = qualified/assembly-level; U = unsupported and reported as such.

Claim	**Key implementation**	**Audit notes**	**Verdict**
S1 experience → new relational structure under scaffolding	attribution pipeline (scoring / aggregation / promotion) + write-back switch	4 degenerate + 2 window-contaminated are self-reported artifacts; code matches	S
S2 persistence and cross-context re-activation	graph save/load + snapshot rotation + shrink guard	4/12 carried by episodic structure (access layer)	S
S3 ledger channel changes action selection	reward/trial ledgers → candidate prior bonus	difference lands at action-selection layer, not completion layer—matches paper's self-scope	S
S4 event framework changes action selection	settlement write-back; `enabled=False` (config)	audit confirms production-default off, zero graph writes	S (assembly)
S5 scalar modulation of propagation dynamics	internal-state tonic/level; modulator as graph citizen; projection	graph is the effect source; dose-response gate ran	S
S6 mechanics tables necessary; zero-prior lock	mechanics tables + goal system	P0 locks at unreachable goal = upstream experience-generation failure; behavior matches code	S
S7 relation direction limits propagation	directed graph; 8 bidirectional whitelist vs forward; recipe edges sinkified	independent re-check ρ .33→.65, r=.886, 10/10; no behavioral test; production graph unmodified	S (mechanism)
Q1 diffusion produces a distinct access pattern (not behavior)	diffusion engine (step/emit)	independent falsification campaign: 663 runs, JCG> 0 0.970	Q
U2 semantic write-backs consumed by planner	planner does not read semantic write-back graph	code audit: no consumption path	U
U5 routing robustly better than baselines	routing harness	520/520 valid; competing-targets raw p=.046, corrected $p\approx1$; tokens 2.1–3.7×	U (directional)
(extra) production answer pipeline renders attention/demand	annotation rendering switch (wired after audit)	audit-identified definitional gap repaired; rendered as projection references, values not exposed	fixed

Table S7: Cells where FAS is strictly better (success@1 over 30 seeds; all at $w \leq 0.5$ and $d \leq 1$).

k	d	w	FAS	flat
2	0	0.4	8/30	0/30
2	0	0.5	30/30	0/30
3	0	0.3	30/30	0/30
3	0	0.4	30/30	0/30
4	0	0.3	30/30	0/30
4	0	0.4	30/30	9/30
6	0	0.3	30/30	0/30
4	1	0.4	9/30	0/30
4	1	0.5	30/30	0/30
6	1	0.3	30/30	0/30
6	1	0.4	30/30	0/30

Table S8: Cells where flat is strictly better (all at $w \geq 0.7$ and $d \geq 1$).

k	d	w	FAS	flat
2	1	0.7	0/30	30/30
2	1	0.8	8/30	30/30
2	2	0.8	0/30	30/30
2	2	0.9	0/30	30/30
3	2	0.7	0/30	30/30
3	2	0.8	0/30	30/30
3	2	0.9	0/30	30/30
4	2	0.7	0/30	30/30
4	2	0.8	9/30	30/30

Table S9: Determinism re-run: agreement with the archived campaign (5 seeds × 4 conditions).

Condition	**Success matches**	**First-action matches**	**Full-sequence matches**
fas_full	5/5	5/5	3/5
D-flat (embedding)	5/5	4/5	4/5
D-noact	5/5	5/5	0/5
llm-direct	5/5	5/5	0/5
Total	20/20	19/20	7/20

Table S10: Routing v2: task-level results (n=20 paired seeds, Wilcoxon; uncorrected p).

Task	**Key contrast**	**Numbers**	**Significance (uncorrected)**
T1 knowledge reuse	FAS vs direct decisions	3.25 vs 4.2	p=0.064 (directional fewer)
T1 knowledge reuse	FAS vs history	3.25 vs 2.5	p=0.035 (worse)
T2 interference suppression	all conditions completion	20/20	none (ceiling)
T3 environmental re-routing	FAS vs history / rag	4.15 vs 2.3 / 2.8	p=0.003 / 0.044: slower
T4 competing targets	FAS vs direct completion	19/20 vs 15/20	p=0.046 (directional)
T4 competing targets	decisions	5.45 vs 7.25	p=0.029
T4 competing targets	target utilization	0.48 vs 0.30	p=0.010

Table S11: Claim → evidence → source mapping (all sources are archived files read programmatically).

Claim	**Evidence**	**Source (archive)**
Beacon: persistent reuse 20/20 vs 0/20	McNemar 2×10^{-6}; first-action 20/20	beacon campaign analysis (80 runs)
Beacon: activation not necessary	D-flat 20/20; decisions Wilcoxon p=0.72	same, + embedding accessibility check (S6)
Write-back is the sole structure source	0 vs 13 learned nodes; 15 promotions (11+2+4)	write-back isolation + promotion taxonomy (S5)
Cross-context re-activation	4/12 vs 0/8 vs 0/0 ($a \geq 0.05$)	re-activation campaign, per-seed records
Ledger channel flips action selection	13 vs 16 ticks; prior (1.0, obs=8) vs (None, 0)	transfer campaign analysis
Event framework flips action selection	@6 vs @9; receipts @15 vs @17	event-framework campaign (assembly)
Convergence mechanism runs	JCG> 0 200/200; hypothesis MRR@1= 1.0	mechanism campaign statistics + 663-run falsification
H3 rejected	ΔJCG $-0.402/-0.402/-0.784/-0.782$; reverse-direction Holm 1.5×10^{-9}; 0/200	mechanism campaign statistics (preregistered)
Shared-hub exception	FAS 30/30 vs 0/30 (single weight)	combinational-topology family
Advantage region mapped	11/84 win, 9/84 loss, 64 tie; O3 30/30 replication	shared-hub sweep `paper_numbers.json` (S8)
Dilution trade-off	ρ -0.445 / -0.656	mechanism campaign dilution sweeps
Direction semantics limit	ρ 0.325→0.645; reach2 0→10/9/12	relation-direction campaign (S9)
Routing: directional only	19/20 vs 15/20; corrected $p \approx 1$; tokens 2.1–3.7×	routing v2 campaign statistics (S11)
Zero-prior starvation	0/25; ledger 0/0/0; +29 nodes	prior-boundary campaign (S12)
Long chain needs task priors	315 / 0-of-5 / 315; −diff 259	long-chain campaign (S12)
Hebbian sub-threshold	3 edges, +0.074; behavior flat	Hebbian isolation campaign (S12)
Affect: access not action	49.8/52.5/57.4; disabled bitwise neutral	affective dose campaign (S12)
Intention expression unattained	peak 0.46 < 0.78	intention lifecycle campaign (S12)
Decision-head determinism	outcomes 20/20, trajectories 7/20	determinism re-run (S10)
Persistence causally operative (context removal)	T2 FAS-full 10/10 *vs* reset 0/10, McNemar p=0.00195; T3 repeats 2.8 *vs* 7.7, Wilcoxon p=0.00098	persistent-state experiment (S16)
Structure-preserving controls: no T2 difference; T3 confounded	C3/C4 10/10 (T2), 12.0/11.7 (T3); failure-evidence visibility; gate-stopped revision, untested	controls experiment + T3 audit (S16)

Table S12: Campaign v2: headline results (n=10 paired seeds unless noted).

Question	FAS	Baselines
Goal resumption (C, closed-book free phase)	**9/10**	history 0/10; direct 0/10 (McNemar p=0.0039)
Resumption, mechanism label	zero-experience control reproduces the same working set	prior-closure edge + observation-triggered activation (not an intention mechanism)
Goal revision (E)	10.5 failed repeats; pivot 7/10 @4.7	history 1.0 repeats; pivot 10/10 @1.0 (FAS worse)
E repair (event-frame polarity wired)	**7.2 repeats**; pivot 9/10 @3.1 (one-sided exact Wilcoxon p=0.0098)	direction confirmed, parity not reached (E1/E3 residual)
Self-model (F)	write path verified (5/5/5 vs 0/0/0); consumer repair: gathers 1.0→6.0	`current_goal` has zero production readers (write-only)
Behavioral convergence (I)	0–2/10 (incl. k=16/32 and rich serialization 0/30)	raw history **10/10** (Holm p=0.041)
Credit horizon (B, zero-LLM, n=10/delay)	≤80 s: 10/10 attributed; 160 s: 0/10	90 s attribution window = the credit horizon
100-cycle stability (J, zero-LLM)	nodes +3.7%, activation clamped	promotions 0 (repetition-starved)
History compression (K)	923 tokens/run, 10/10	history-200 13,290 tokens/run, 10/10

Table S13: Persistent-state advantage experiment (n=10 paired seeds).

Task	FAS-full	Baselines
T2 goal-completion	**10/10**	reset 0/10 (McNemar p=0.00195); direct 0/10; h128/h512/h2048 10/10, 10/10, 9/10; RAG 10/10
T3 failure-informed	repeats **2.8**; pivot 11/11	reset repeats 7.7 (Wilcoxon p=0.00098); direct 11.2; h2048 0.0; RAG 0.0
T1 hunger (sanity)	hunger restored 11/11	all conditions 10–11/11 (ceiling)
FAS-sham (nodes, no edges)	10/10 on T2	≈ FAS-full (node traces + gate suffice)

Table S14: Controls: flat-store and graph-no-activation (n=10 paired seeds).

Condition	T2 success	T3 mean fails	Interpretation
C1 FAS-full	10/10	2.8	reference
C2 FAS-reset	0/10	7.7	no persistent state
C3 Flat-Store	10/10	12.0	persistence without graph structure
C4 Graph-NoAct	10/10	11.7	graph without spreading
C1 vs C3 (T3)	—	—	HL=−9, CI[−11,−7.5], Bonf p=.0156: C1 better
C1 vs C4 (T3)	—	—	HL=−9, CI[−10,−7.5], Bonf p=.0156: C1 better
C3 vs C2 (T3)	—	—	HL=+4.5: C3 worse than reset
C4 vs C2 (T3)	—	—	HL=+4.0: C4 worse than reset

Table S15: Statistical conventions, full version.

Family	**Test**	**Sidedness**	n	**Correction (m)**	**Status**	**Implementation (archived)**
Beacon success	exact McNemar	two-sided	20	none	pre	`scripts/analyze_beacon.py`
Beacon decisions	paired Wilcoxon	two-sided	20	none	pre	`analyze_beacon.py` (default two-sided)
Mechanism ΔJCG	Wilcoxon, normal approx. (analytic tail)	one-sided; reported Holm = reverse tail; prereg tail ≈1	50	Holm (4)	pre	`core_incremental_value/runners/` (frozen; step 3b)
Mechanism secondary	paired Wilcoxon	two-sided	50	none	pre	same `analyze.py`
Shared-hub sweep	McNemar; Wilcoxon	two-sided	30	none	pre	`shared_hub_sweep/extract.py`
Relation direction	Wilcoxon (exact)	two-sided	10	single family (21)	pre	`scripts/run_exp_reldir.py`
Campaign v2 C	exact McNemar	two-sided	10	Holm (exp.)	pre	`capability_gap_v2/analyze_campaign.py`
Campaign v2 E2/T	Wilcoxon (exact)	one-sided	10 / 9	Holm (exp.)	pre	raw `raw_E/raw_T` + audit recomputation (P_VALUE_AUDIT #6/7)
Campaign v2 I	paired Wilcoxon	two-sided	10	Holm (exp.)	pre	`analyze_campaign.py`
Campaign v2 B	descriptive counts	—	10	—	pre	`delayed_credit.py`
Controls C3/C4	McNemar; Wilcoxon	two-sided	10	Bonferroni (8)	pre	`persistent_state_controls/analyze_controls.py`
Persistent state H1/H3	exact McNemar	two-sided	10	none	pre	`persistent_state_advantage/analyze_psa.py`
Persistent state H5	Wilcoxon	one-sided (greater)	10	none	pre	`analyze_psa.py`
Routing	paired Wilcoxon	two-sided	20	Bonferroni (280)	post	`scripts/analyze_routing_v2.py` (step 5)
Cross-episode	McNemar; Wilcoxon	two-sided; one-sided (greater)	30	within exp.	pre	`cross_episode_recomb/analyze.py`
Determinism re-run	agreement counts	—	20	—	desc	`beacon_rerun/compare.py`
Presentation-equivalence revision	none (gate-stopped, 0 runs)	—	0	—	post	`persistent_state_controls_revision/`